Six clues in one shifter click

How Bike Derailleurs Work

Name _____ Date _____



Little traced the chain without touching it. The loop wrapped around a front chainring and one cog in the back cassette. Each gear had teeth that fit between the chain's rollers. When the pedals turned forward, the chain carried that turn toward the back wheel.



Big pressed the shifter once while the chain moved. The lever pulled a short length of cable. That pull moved the rear derailleur sideways. When the shifter let cable out, a spring drew the derailleur back the other way.



Two small wheels turned inside the derailleur cage. The upper guide pulley lined the chain up with a cog. The lower tension pulley moved as the spring turned the cage. That motion took up extra chain, called slack, so the long loop stayed snug in every gear.

gear	a toothed wheel that helps transfer a turn
chain	linked metal pieces that carry a pedal turn to the back wheel
cog	one toothed gear in the stack on the back wheel
teeth	the points on a gear that fit the chain
derailleur	a bike part that guides the chain sideways between gears
cable	a thin wire that carries a shifter's pull
spring	a part that stores a pull and moves the derailleur back
lever	a handle that moves when someone presses or pulls it
shift	move the chain from one gear to another
easy / hard gear	less pedal push with fewer wheel turns / more pedal push with more wheel turns

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This bike had two chainrings in front. A second derailleur sat above them. Its narrow cage surrounded the chain. As the cage moved sideways, it pressed the moving chain off one chainring so the neighboring chainring could catch it.



Little noticed that a click did not lift a still chain onto new teeth. The pedals had to turn forward. Then each arriving chain link could meet the shaped teeth on the next cog and climb into place. A derailleur guides; the moving chain completes the shift.



On the biggest back cog, one pedal turn produced fewer wheel turns and needed less push. That easy gear suited a hill. On a small back cog, the wheel turned farther for each pedal turn, making a harder gear useful on flat ground. Tooth counts tell the ratio: 32 front teeth divided by 16 back teeth equals 2 wheel turns for one pedal turn.

1. How do the cable and spring move the rear derailleur in opposite directions?
2. Why does the rear derailleur need both a guide pulley and a tension pulley?
3. How does the evidence about tooth counts explain two wheel turns from a 32-tooth chainring and a 16-tooth cog?

Your turn

Write one sentence that follows a shifter click to the chain.

Example: The lever pulls the cable, so the rear derailleur guides the moving chain sideways.